ASSESSMENT TOOL 2

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AI-Based Elderly Companion Robot with Intelligent Medication Reminder and Health Monitoring System

Abstract:

The increasing elderly population has created a growing demand for intelligent healthcare solutions that support independent living. Many senior citizens forget to take medicines, miss appointments, or face medical emergencies while living alone. This proposed AI-based companion robot provides personalized medication reminders, monitors vital signs, detects falls, and communicates with caregivers during emergencies.

Problem Statement:

• Elderly people often forget medications and appointments.

• Traditional reminder systems only generate alarms and cannot verify medication intake.

• Lack of continuous health monitoring increases emergency risks.

• Need for an intelligent, adaptive, and healthcare assistant.

Latest AI Application:

It represents a modern AI healthcare application by combining:

• Machine Learning

• Natural Language Processing (NLP)

• Expert Systems

• IoT-enabled Health Monitoring

• Edge AI for decision making

AI Technologies Used:

• Machine Learning: Learns daily routines and predicts health risks.

• Natural Language Processing (NLP): Enables voice communication between the robot and the elderly.

• Expert System: Uses medical knowledge rules to generate healthcare recommendations.

• IoT : Monitor heart rate, body temperature, oxygen level, and movement.

• Cloud Computing: Stores health records for long-term analysis.

Working Mechanism:

1. Health sensors collect vital signs.
2. Data is processed and cleaned.
3. ML model helps in health conditions.
4. Expert System applies medical rules.
5. Robot reminds users to take medicines.
6. NLP allows natural voice interaction.
7. In emergencies (fall or abnormal vitals), caregivers receive instant alerts.
8. Health data is securely stored in the cloud.

Literature Review:

Recent research highlights:

• AI improves medication adherence and patient monitoring.

• IoT-based system helps enable continuous health tracking.

• NLP enhances elderly–robot interaction.

• Expert Systems provide explainable medical recommendations.

• Current systems lacks personalized decision-making, which this proposed system addresses.

Real-World Implementation, Impact & Challenges:

Example:

An elderly patient living alone forgets morning medication. The robot provides a voice reminder. If no response is detected and health sensors detect an abnormal heart rate or a fall, the system automatically alerts family members.

Applications:

• Smart Homes

• Hospitals

• Assisted Living Centers

• Rehabilitation Centers

Advantages:

• Personalized medication reminders

• Continuous health monitoring

• Early emergency detection

• Reduced caregiver workload

• Improves safety

• Supports independent living

Challenges

Future Scope:

• Integration with wearable devices

• Emotion recognition using AI

• Predictive disease diagnosis

• Generative AI for natural conversations

• Smart home automation integration

Originality:

This robot**:-**

• Learns user habits using Machine Learning.

• Uses NLP for natural communication.

• Combines Expert Systems with IoT for explainable healthcare decisions.

• Provides real-time emergency detection and healthcare assistance.

Conclusion:

The AI-Based Elderly Companion Robot is a modern intelligent healthcare solution that combines AI, ML, NLP, Expert Systems, and IoT to improve medication adherence, monitor health conditions, detect emergencies, and support independent living. The proposed system ensures patient safety, reduces caregiver burden, and represents a future AI-driven elderly healthcare.